OligoTox-CNS — Methodology

Materials and methods for the CNS oligonucleotide toxicity dataset
NIH/NCATS Oligonucleotide Toxicity Open Data Challenge — Phase 2 · release v1.0

1 Design

OligoTox-CNS is a **curated** dataset. No new wet-lab experiment was performed; the materials and methods below are therefore of two kinds, and are kept strictly separate: **(a)** the experimental methods of the source studies, recorded verbatim so that a user knows how each measurement was generated, and **(b)** the curation methods used here to extract, verify and harmonise them. Conflating the two would let curation choices masquerade as experimental fact.

2 Source selection and retrieval

Inclusion required a CNS-specific toxicity outcome attributable to an identified oligonucleotide. Sources were ranked by whether they also published the sequence. Excluded: sources whose toxicity endpoints were hepatic, renal or systemic-immune without a CNS readout; and any value reachable only through a web-search summary.

Supplementary files were retrieved through the Europe PMC REST supplementary-files endpoint (`/europepmc/webservices/rest/<PMCID>/supplementaryFiles`). This route was adopted after the PubMed Central interface was found to gate binary downloads behind a JavaScript proof-of-work challenge that returns an HTML stub in place of the requested file — a stub that would otherwise be silently parsed as data. Regulatory labels were read from DailyMed, identified by set ID and label publication date. Every retrieved file is committed to `sources/`.

3 Oligonucleotide identity, purification and characterisation

> This section answers the challenge's requirement for *the methods used to purify and characterize oligo identity*. Because the compounds were synthesised by the source laboratories and not here, what can be reported is what each source states. **Where a source is silent, the field is `NOT_REPORTED`. No purity value has been estimated, inferred from a synthesis platform, or carried across from another compound.**

Reported by source H1, covering 1,825 of the 1,879 oligonucleotides

Quoted verbatim from the source's Materials and Methods: *“Single-stranded DNA oligonucleotides with complete PS backbones and LNA-modified flanks were synthesized on a MerMade 192× synthesizer (Bioautomation, TX, USA) following standard phosphoramidite protocols. The final 5′-dimethoxytrityl (DMT) group was left on the oligonucleotide for later use as lipophilic handle and chromatographic retention probe. The oligonucleotides were purified by solid phase extraction in TOP cartridges (Agilent Technologies, Glostrup, Denmark) using the DMT group, after which the DMT group was removed … The oligonucleotides were dissolved in sterile 0.9% saline solution, and the concentration of oligonucleotide in solution confirmed by calculating the Beer-Lambert extinction coefficient and measuring ultraviolet absorbance. Oligonucleotide identity and purity were validated by reversed-phase ultra-performance liquid chromatography coupled to mass spectrometry.”*

These statements populate `synthesis_platform`, `purity_method`, `identity_confirmation` and `formulation`. **The source states the method but publishes no per-compound purity percentage and no observed mass**, so `purity_pct` is `NOT_REPORTED` for every row.

Reported by the remaining sources

Source	Synthesis	Purification	Identity confirmation	Purity value
K1	not reported	not reported	not reported	not reported
L1	commercial synthesis stated; platform not reported	not reported	not reported	not reported
C1	not applicable — approved products manufactured to their marketing authorisation	not in label	not in label	not in label

Table M1. *Characterisation reporting by source. Across the whole release, 1,859 of 1,879 oligonucleotides carry a stated purification method and* ***0 carry a purity value****. This is a property of the published literature; see* `OPEN_ITEMS.md` *OI-02.*

Sequence identity as recorded here

Identity in this dataset means the printed sequence together with its per-position chemistry. Both are transcribed mechanically, never typed by hand, and both are checked against internal evidence: for all 1,825 H1 compounds the count of upper-case characters in the printed sequence equals the source's own declared LNA count (0 mismatches), and every oligonucleotide that has a published sequence has a declared length and base composition equal to that sequence (1,858 of 1,858; the remaining 21 have no published sequence to check against). Both are re-checked by the QC suite on every build.

4 Experimental methods of the source studies

Source	System	Design
H1 — in vivo	adult female C57BL/6J mice	Single intracerebroventricular bolus, 100 µg in 5 µL 0.9% saline. Observed 1 h using a modified functional observational battery per the Oligonucleotide Safety Working Group. Five categories (hyperactivity; decreased activity and arousal; motor dysfunction/ataxia; abnormal posture and breathing; tremor/convulsions), each 0–4, summed to 0–20. 4–6 mice per group; per-animal scores averaged to a group score.
H1 — in vitro	primary cortical neurons, Sprague-Dawley rat embryonic day 19, 25,000 cells/well	Spontaneous calcium oscillations, fluo-4 AM, read on FLIPR for 300 s, ± 1 mM added Mg^{2+}. ASO at 25 µM, chosen as the expected CSF concentration in the first hour after a 100 µg ICV injection. Score = 1 point per 1-s read whose signal increase exceeds 50% of mean control amplitude, summed and expressed as percent of control.
K1	wild-type FVB and two Huntington's disease model mouse lines	Bilateral ICV injection of di-siRNA and ASO constructs, 1.25–20 nmol, in vehicles containing 0–32 mM Ca^{2+} and/or 0–16 mM Mg^{2+}. Acute tolerability scored on the same 0–20 scale; 4–6 animals per group.
L1	C57BL/6 and ICR mice; Slc:SD rats	ICV injection of 15.2–39.9 nmol, or intrathecal injection via spinal canal catheter of 190 nmol. Five-category 0–4 tolerability scale (separate rat variant), open-field locomotion and body weight, assessed to day 21. n = 4 per group.
CT1	human patients, 22 randomised trials	Adverse-event tables as posted to ClinicalTrials.gov by the sponsors, covering trials of nusinersen, tofersen, tominersen, BIIB080/MAPTRx, BIIB105, WVE-120101, WVE-120102 and WVE-003 delivered intrathecally or intracerebroventricularly. Each event is reported per arm with a numerator (patients affected) and a denominator (patients at risk), including comparator arms. Exposure spans months to years of repeat dosing.
C1	human patients	Randomised placebo- or sham-controlled trials. Tofersen 100 mg intrathecally (n = 72 versus 36 placebo); nusinersen 12 mg intrathecally (n = 84 versus 42 control in the later-onset study). Adverse-event incidences as printed in the prescribing information, plus post-marketing reports with no estimable frequency.

5 Extraction and transcription

- **Spreadsheet source (H1).** Read cell-by-cell with `openpyxl`. No value passed through a language model or a summarising fetch layer, which was a deliberate decision after a transcription artefact (a Cyrillic character substituted into a nucleotide sequence) was observed in summarised text during source discovery.
- **Typeface-encoded source (L1).** Chemistry is carried in **bold** = LNA and ***bold-italic*** = 2′-MOE, with `C(5)` marking 5-methylcytosine; plain text extraction discards all of it. Per-position chemistry was recovered by reading PDF span style flags (`src/build_curated.py::parse_kuroda_sequences`), yielding 3-10-3 and 4-12-4 LNA gapmers and one 5-10-5 2′-MOE gapmer.
- **PDF table source (K1).** Parsed by a state machine over the text layer, keyed on runs of six consecutive numeric fields, recovering 41 dosing groups. Curves were selected by the source figure panel each group belongs to, because selecting by dose and cation alone merges control groups from unrelated panels.
- **Regulatory source (C1).** Warnings, incidences and denominators read directly from the label text.
- **Trial registry source (CT1).** Retrieved through the ClinicalTrials.gov API v2 (`resultsSection.adverseEventsModule`), not the results web page — that page is a client-side application whose HTML contains none of the adverse-event text, so scraping it returns nothing while appearing to succeed. Ingestion is restricted to CNS-relevant events: the whole MedDRA *Nervous system disorders* class, plus a curated set of terms filed under other classes that are nonetheless CNS events (post-lumbar-puncture syndrome, CSF findings, meningitis, myelitis, papilloedema, hydrocephalus). Cardiac “ventricular” terms are excluded — those are heart ventricles. Each trial's investigational oligonucleotide is assigned from an explicit per-trial table rather than parsed from free text, after parsing produced junk identifiers that would have split one molecule across several

rows. The retrieved JSON is committed, so the build does not depend on the service remaining available.

6 Harmonisation and grading

Sources H1 and K1 report on the same 0–20 acute tolerability scale and are graded identically. The 0–3 ordinal grade uses the cut-offs published by the H1 authors (4, 7, 18), not thresholds devised here: grade 0 = score 0; grade 1 = 0–4 ('mild'); grade 2 = 4–7 ('moderate'); grade 3 = above 7. The boundary between grade 1 and 2 is the authors' own developability line, and it reproduces: 112 of 181 in vivo rows (61.9%) fall at grade ≤ 1, against the authors' stated 'roughly 60%'. Every graded row carries the rule applied in `grade_basis`, and every grade is `provisional`.

Where a source publishes scores only as figures, no number is read off the figure: `readout_value` is `NOT_REPORTED`, `readout_is_qualitative` is TRUE, and the grade is taken from the severity the authors state in words. Clinical rows use a separate rubric and are kept on separate `tox_axis` values so that they are never silently pooled with preclinical scores.

7 Quality control

`qc/validate_dataset.py` runs 34 checks; all 34 pass and the script exits non-zero on any failure. They cover: primary-key uniqueness; referential integrity across all three joins; controlled-vocabulary conformance on seven columns; grade range; the requirement that every graded row state its grading rule; the requirement that **every numeric readout name its source table or figure**; agreement between each declared length and base composition and the actual sequence; and, for every oligonucleotide, that the modification table holds exactly one contiguous row per position whose nucleobase matches the sequence at that position.

Two independent end-to-end checks were also run. The source authors' published predictive model, re-implemented from their supplementary methods, reproduces their own score column for all 1,825 rows. And a baseline classifier trained only on the released CSVs, using the source's own held-out split, reaches 89.5% accuracy — matching the accuracy the authors report for that validation set.

8 Reproducibility

The pipeline is deterministic and fully committed. From a clean checkout: `build_hagedorn.py` → `build_curated.py` → `assemble.py` → `validate_dataset.py` → `make_figures.py` → `make_release.py` → `make_pdfs.py`. Dependencies are `openpyxl`, `pymupdf`, `matplotlib` and `reportlab`. Every number in the figures and in both PDFs is computed at build time from `data/`, so neither document can state a value the dataset does not contain.

Determinism is **byte-level, and checked rather than asserted**. Running the pipeline twice over the same inputs reproduces all 19 released artefacts — the four CSVs, the workbook, both PDFs, the eight figures and the generated documentation — with identical SHA-256 digests. Reaching that required suppressing three sources of build-time noise that would otherwise make every rebuild differ while the data stayed the same: ReportLab's embedded creation timestamps and document IDs, the archive timestamps inside the .xlsx container, and the modification date openpyxl writes into `docProps/core.xml` at save time. A reviewer who re-runs the build can therefore diff the outputs against the committed ones and expect an exact match, rather than having to compare them by eye.